

Numerical simulations of biaxial test using non-spherical particles

A.A. Peña¹, A. Lizcano¹, F. Alonso-Marroquin² and H. J. Herrmann²

¹ *Universidad de los Andes, Bogotá D.C., Colombia*

² *ICP-University of Stuttgart, Stuttgart, Germany*

ABSTRACT: The mechanical response of cohesionless granular materials under monotonic load is studied by performing molecular dynamic simulations. The diversity of shapes of soil grains is modeled by using randomly generated convex polygons as granular particles. Results of the biaxial test obtained for dense and loose media show that samples achieve the same void ratio at large strains independently of their initial density state. This limit state resembles the so-called critical state of soil mechanics, except for some stress fluctuations, which remain for large deformations. These fluctuations are studied at the micro-mechanical level, by following the evolution of the coordination number and the fraction of the sliding contacts of the sample. The effects of grain friction and the interlocking of grains on the global friction are studied by correlating the macro-mechanical angle of friction with the interparticle friction coefficient. Furthermore, the so-called Swept Out of Memory behavior is also obtained as the asymptotical behavior under proportional strain paths.

1 INTRODUCTION

The stress-strain response of granular materials shows a nonlinear and irreversible behavior when they are subjected to external load. This macro-mechanical behavior stems from the discrete character of the media and phenomena occurring at the grain scale. There are numerous constitutive theories that consider the media as a continuum, but they often involve many parameters that generally lack of physical meaning (Bolton 2002).

Numerical simulations have become a valuable tool in the study of different phenomena occurring at the micro-mechanic scale in granular materials (Cundall et al. 1979; Tillemans and Herrmann 1995). The mechanical response of the media is obtained by modeling the particle interactions as a dynamic process and using simple mechanical laws in these interactions.

In this work the molecular dynamic technique is employed to simulate the biaxial test of discrete media. The grains are represented by randomly generated convex polygons to include the effect of the diversity of shapes of a realistic soil.

Generally, discrete models use discs or spheres to reduce the time of calculation; however, this kind of models does not consider the effect of particle shape on the mechanical behavior. Several researchers have studied this influence; for example, Holubec and D'Appolonia (1973) concluded that the variation order of the mechanical properties due to the shape of the particle, could be of the same magnitude as the variation due to the changes in density. By means of numerical simulations, Mirghasemi et al. (2002) demonstrated that the particle angularity has an important effect in the compressibility and shear strength of the granular media. It is therefore important to take into account the particle shape in order to have a more realistic soil representation.

The outline of this paper is as follows. The discrete model and the sample specifications are introduced in Section 2. Results of the biaxial test simulations are presented in Section 3 from the macro-mechanical and micro-mechanical point of view. Finally, in Section 4 conclusions and recommendations are exposed.

2 MODEL

The random generation of the convex polygons used in this model is carried out by means of a Voronoi tessellation. This is a simple method to discretize the media without introducing any kind of anisotropy (Lauritsen et al. 1993). Once the sample has been created each particle is subjected to interparticle forces and boundary forces. It is assumed that particles interact elastically with each other. The polygons can neither be broken nor deformed, but they can overlap when they are pressed against each other. This overlap represents the local deformation of the grains. This approach has been thoroughly used to model many different processes, such as strain localization and earthquakes (Tillemans and Herrmann 1995), fragmentation (Kun and Herrmann 1996), and damage (Kun and Herrmann 1999).

When all forces F_i acting on the particle i , F^c from interparticle contacts and F^b from boundaries are known, the evolution of the position $\bar{x}_i$ and orientation φ_i of the i polygon is given by the integration of Newton's equation of motion:

$$\begin{aligned} -m_i \ddot{x}_i + \sum_c \bar{F}_i^c + \sum_{c_b} \bar{F}_i^b &= \vec{0} \\ -I_i \ddot{\varphi}_i + \sum_c \bar{I}_i^c \times \bar{F}_i^c + \sum_{c_b} \bar{I}_i^b \times \bar{F}_i^b &= \vec{0} \end{aligned} \quad (1)$$

where m_i denotes the mass of particle i , I_i its moment of inertia and $\bar{l}$ the branch vector which connects the center of mass of the polygon to the application point of the contact force. Each particle has two linear and one rotational degrees of freedom. In the following section the laws of the contact forces are introduced.

2.1 Contact law

The contact force results from the overlap area a between particles. In Figure 1 the configuration of a particle contact is presented, P_1 and P_2 represent the intersection points between the edges of the polygons; the segment that connects those points gives the contact line $\vec{s} = \vec{P_1 P_2}$. This vector defines a coordinate system at the contact $(\hat{n}, \hat{t})$; where $\hat{n}$ and $\hat{t} = \vec{s}/|\vec{s}|$ normal to it give the direction of the normal F_n and tangential F_t components of the contact force. The point of application of the contact forces is taken as the center of mass of the overlap area (Alonso-Marroquín and Herrmann 2002).

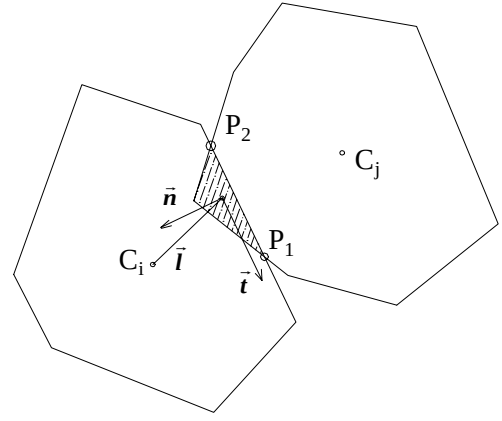


Fig 1. Particle contact configuration

The contact force is calculated as:

$$\vec{F}^c = -k_n \delta \hat{n} - k_t \xi \hat{t} \quad (2)$$

where k_n and k_t are the stiffnesses at the contact in the respective directions. The deformation length is $\delta = a/|\vec{s}|$. Here a is the overlapping area of the interacting polygons. The tangential force is introduced by an elastic spring whose displacement ξ (Eq. 3) is equal to the time integral starting at the beginning of the contact

$$\xi(t) = \int_0^t \bar{v}_t^c(t') \Theta(\mu F_n - |F_t|) dt' \quad (3)$$

where Θ is the Heaviside function, and $\bar{v}_t^c$ the tangential component of the relative velocity $\bar{v}^c$ at the contact.

$$\bar{v}^c = \bar{v}_i - \bar{v}_j - \bar{\omega}_i \times \bar{l}_i + \bar{\omega}_j \times \bar{l}_j \quad (4)$$

here $\bar{v}_i$ is the linear velocity and $\bar{\omega}_i$ is the angular velocity of the particles in contact. The tangential displacement ξ at the contact increases until the Coulomb's limit condition $F_t^c = \mu \cdot F_n^c$ is reached. At this moment the contact slips and the permanent deformation begins.

Finally, we introduce a viscous force, which is necessary to maintain the numerical stability of the method and to obtain a quick convergence to the equilibrium configuration (Eq. 5).

$$\vec{F}_v^c = -m (\gamma_n \cdot \bar{v}_n^c \cdot \hat{n}^c + \gamma_t \cdot \bar{v}_t^c \cdot \hat{t}^c) \quad (5)$$

where $m = (1/m_i + 1/m_j)^{-1}$ is the effective mass of the two particles in contact, and γ_n and γ_t are the damping coefficients. They are related to the normal restitution coefficient during collisions (Tillemans and Herrmann 1995).

2.2 Boundary conditions

During the simulation, the granular sample is confined by four rigid walls. These walls always maintain their horizontal or vertical orientation. Boundary forces are applied on each grain in contact with these walls. The walls are frictionless, so they transmit only normal forces to the polygons in contact with them. When one of the vertices of a polygon penetrates one of the walls, a force, proportional to the penetration length δ , is applied on the polygon. This boundary force $\bar{F}_n^b$ is oriented in normal direction $\hat{n}$ to the wall (Eq. 6).

$$\bar{F}_n^b = k_n \delta \hat{n} \quad (6)$$

$$\bar{F}_v^b = -m_i \gamma_n \bar{v}^c \hat{n} \quad (7)$$

Viscous forces $\bar{F}_v^b$ in wall-polygons interactions are also considered (Eq. 7). Where m_i is the mass of the particle in contact with the boundary wall, and γ_n the damping coefficient in normal direction. The boundary force is calculated for all the cases of interaction between walls and polygons in the same way. Finally, the displacement of the walls and the total force on them are used to determine the global stress and strain of the assembly.

2.3 Parameters of the model

The parameters used in the model are presented in Table 1. The interparticle friction coefficient was chosen $\mu = 0.55$, which yields similar values of macro-mechanic friction angles to those obtained in real materials as gravel or sands (Bardet 1997). The stiffness ratio $\zeta = k_t/k_n$ as well as the viscosity ratio (γ_t/γ_n) are taken 0.33. This value was chosen in order to obtain the Poisson ratio of the real materials (Kruyt and Rothenburg 1996). The normal stiffness k_n was taken as 160 MPa; this value has been adjusted to the Young's modulus of the material (Alonso-Marroquín and Herrmann 2002). The viscosity factors were selected to preserve the stability and to reduce the acoustic waves produced during loading.

Finally, the time ratio in Table 1 is the ratio between the time of loading t_o and the characteristic period of oscillation $t_s = \sqrt{m_0/k_n}$ where m_0 is the mean mass of the polygons. The value of t_s determines the selection of the time step Δt . In order to preserve the stability of the method, it should be smaller than $0.25 t_s$.

Table 1. Parameters used in the model

Parameter	Ratio	Value
Friction coefficient	μ	0.55
Stiffness ratio	$\zeta = k_t/k_n$	0.33
Viscosity	γ_t/γ_n	0.33
Time ratio	t_o/t_s	1000
Time step	Δt	$2.5 \cdot 10^{-6}$ s

2.4 Sample construction and specification.

After generation of the polygons, the sample is confined by applying a gravitational field to the particles and on the walls, oriented to the center of mass of the assembly. Finally, the sample is subjected to an isotropic compression until the desired confining pressure is reached. In order to generate dense samples, the interparticle friction is set to zero during the construction. The loose samples are created taking damping coefficients 100 times greater than those used in the test stage. In both cases, the parameters are set to the values shown in Table 1 after the preparation of the sample. The sample specifications used to perform the simulations were: 1000 kg/m³ for the density of particles, a sample size of 400 polygons, and a unitary shape ratio (width/height).

The average area of the polygons could be considered unitary; therefore the particle's mean diameter can be taken as ≈ 0.01 or 0.001 m. Figure 2 shows the particle size distribution curve of the virtual samples, corresponding to the two values of mean diameter. The soil uniformity and the particle size distribution can be characterized by the uniformity coefficient C_u and the coefficient of curvature C_c , respectively. These coefficients are defined in Eq. (8) and they are given in Table 2 for the average grading of the samples.

$$C_u = \frac{D_{60}}{D_{10}} \quad (8)$$

$$C_c = \frac{D_{30}^2}{D_{10} D_{60}}$$

where D_{10} , D_{30} and D_{60} are the grain size corresponding to 10, 30 and 60% finer, respectively (Fig. 2). According to the unified soil classification system USCS (ASTM D-2487-69), and the coefficients C_u and C_c the virtual samples are classified as poorly graded gravel (GP) or sand (SP).

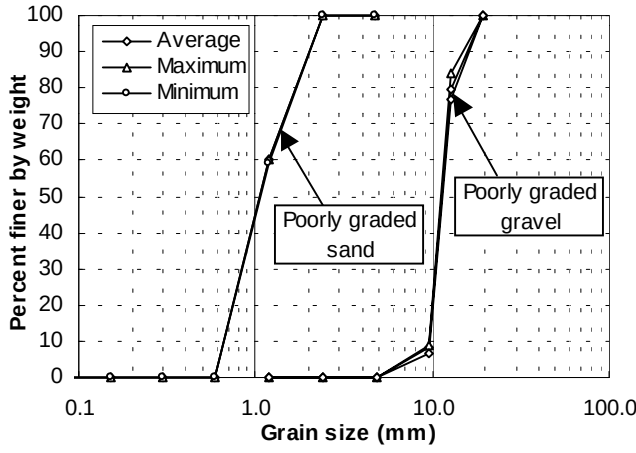


Fig. 2. Grain size distribution chart from virtual samples

Table 2. Coefficients of uniformity and curvature for virtual samples.

Particle's mean diameter	Coefficients		USCS
	C_u	C_c	
0.01 m	$1.22 < 4$	$0.96 < 1$	GP
0.001 m	$1.81 < 6$	$0.89 < 1$	SP

3 BIAXIAL TEST SIMULATIONS

Two different types of biaxial test simulations were performed. In the first one, the sample is loaded using strain control on the horizontal walls (axial direction), while the stress on the vertical walls remains constant. The stress-strain behavior at the macro-mechanical scale, for dense and loose systems, and its dependence on some mechanisms at the micro-mechanical scale are discussed. The effect of the interparticle coefficient of friction on the macro-mechanical response of the media is also studied. In the second test, the sample undergoes proportional strain paths in order to reproduce the so-called Swept Out of Memory (SOM) behavior of granular materials.

In the tests the axial and lateral directions are indicated as 1 and 3, respectively. That is σ_1 and ϵ_1 are the axial stress and strain, and σ_3 and ϵ_3 are the lateral components. Stresses have the same sign convention used in soil mechanics: compressive normal stresses are positive and tensile normal stresses are negative. The stress values are normalized by the normal stiffness at the contacts ($k_n = 160$ MPa).

3.1 Test 1: Shear simulations

The parameters used to perform this type of biaxial tests are presented in Table 3.

Table 3. Parameters of the biaxial test.

Parameters	Value
Default confining pressure	160 kPa
Axial strain rate	0.02 sg^{-1}
Maximum axial strain	20 %

3.1.1 Macro-mechanical observation

Three different samples were used to evaluate the mechanical response of polygonal packings; each one corresponds to a different seed used in the random generation. The initial void ratios of the dense and loose samples are presented in Table 4.

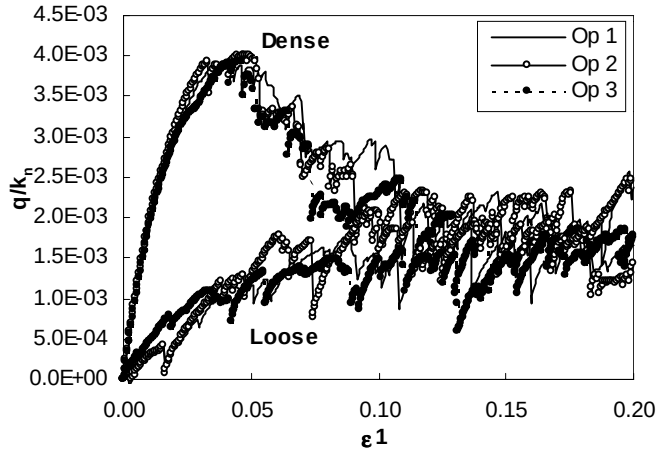
Table 4. Sample initial void ratio

Sample	Void ratio	
	Dense state	Loose state
1	0.128	0.251
2	0.130	0.262
3	0.135	0.271

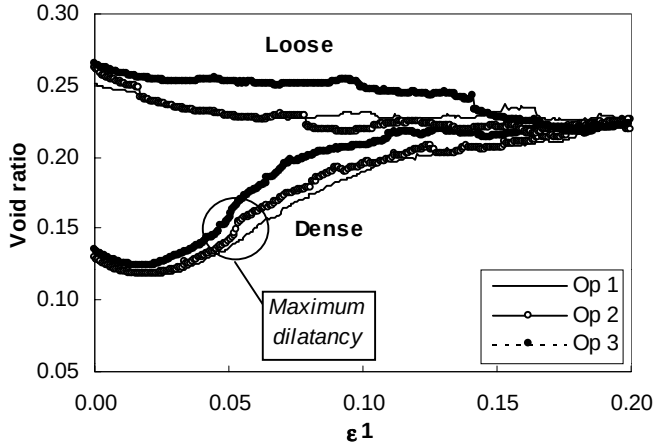
Figure 3(a) shows the evolution of the deviator stress q ($\sigma_1 - \sigma_3$) with axial strain ϵ_1 for the dense and loose system. In general, the dense samples exhibit a higher initial stiffness modulus than the loose ones. After the peak in the dense media, which is about 5% axial strain, a strain-softening behavior is observed. The loose media exhibit more frequent and bigger variations in the stress behavior. Additionally, a peak strength is not observed. Although an increase of fluctuations of the stress are observed for both systems, it presents a tendency to stabilize around a value that one could consider as a steady state of the material ($dq/dt = 0$).

The evolution of the void ratio with axial strain is illustrated in Figure 3(b). Initially, the dense samples contract and later the void ratio increases (dilatancy). For large axial strain values the void ratio reaches a constant value. Comparing the evolution of dense samples in Figure 3 (a) and (b) we notice that the maximum rate of dilatancy agrees with the peak strength ($\approx 5\%$ axial strain) which is expected on soils. It is especially observed in samples 2 and 3.

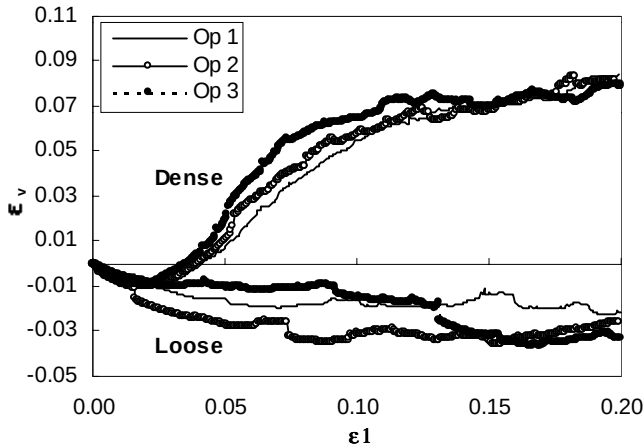
The loose samples reduce their void ratio during the test (Fig 3(b)), and it tends to a constant ratio near to 18% axial strain. The void ratio in both dense and loose samples varies until it achieves the same value near to 0.23 corresponding to 18% axial strain. In this stage, the granular medium is deformed at constant volume and with the same approximate value of deviator stress, which corresponds to the critical state of the material and it is independent of the initial sample density (Schofield and Wroth 1968).



(a)



(b)



(c)

Fig. 3. Evolution of (a) the deviator stress (b) void ratio and (c) volumetric strain during the biaxial test with 400 particles.

The volumetric strain evolves, increasing in the dense system and decreasing in the loose one (Fig 3(c)). At large values of axial strain the samples tend to deform at constant volume. All these features reproduce the asymptotic behavior of soils obtained in laboratory experiments.

3.1.2 Evolution of internal variables

By means of numerical simulations it is possible to examine how the internal variables of the system evolve. The average number of contacts per particle within the media is defined as the coordination number γ . The evolution of the coordination number of one of the samples is presented in Figure 4.

At low axial strain values, the dense system contracts and as a consequence a small increment of γ is observed. This is followed by a decrease of the γ value when system dilates. This decrease is associated to the tendency of the particles to break the interlocking among them during the shear process. As a result, each particle begins to lose contacts. This is macro-mechanically observed by the trend of dense samples to increase their volume. In contrast, the loose granular sample tends to a denser structure; therefore new contacts are generated.

Both samples around 8% axial strain attain the same critical value of γ , which is independent of the expansion or contraction of the medium. This critical average coordination number could be related to a fundamental physical requirement of stability of the media (Thornton 2000).

Figure 5 shows the evolution of the ratio between the number of sliding contacts and the total number of contacts. The sliding condition is given by the Coulomb's condition, $F_t^c = \mu \cdot F_n^c$. Initially, the dense medium has more sliding contacts when it contracts. Later when the sample begins to expand the number decreases. In general, the evolution of the sliding contacts for both systems during loading, consists of stages where their number increases, and short time "failures" where the fraction of sliding contacts jumps down.

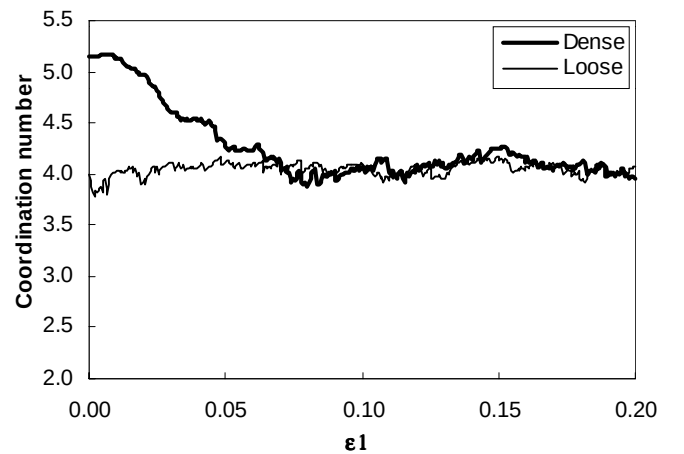


Fig. 4. Evolution of the average coordination number during the biaxial test

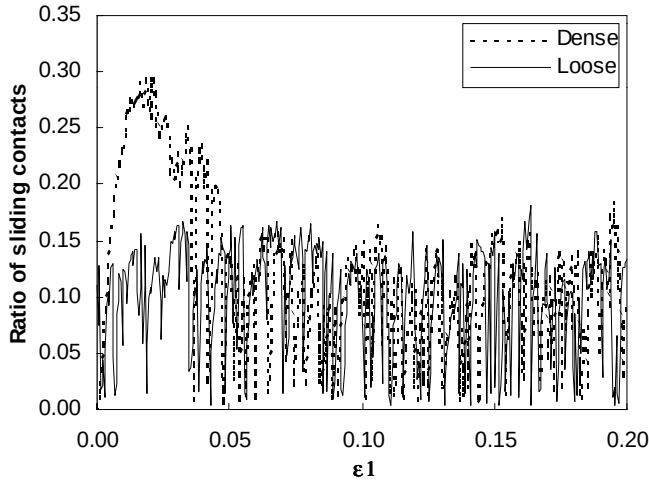


Fig 5. Evolution of the fraction of sliding contacts ratio during the biaxial test

Figure 6 compares the stress-strain evolution to the fraction of sliding contacts for the dense and loose sample. We observe more initial stability, with low frequency of jumps in the stress, in the dense sample. This stability is related to the average coordination number of the medium (Fig. 4), and the bigger this value the bigger the resulting stability of the granular skeleton. Although the jumps observed in the stress-strain behavior are less frequent than ones in the sliding contacts (Fig. 6), each stress jump is associated to an abrupt reduction of the number of sliding contacts.

If one performs a detailed analysis of the evolution of these variables, one finds that each stress-strain fluctuation matches with a fluctuation of the fraction of sliding contacts at the same Δt . It seems to be also confirmed by the study of the displacement vectors of the individual particles (Peña 2004). Where big displacements of the particles are associated to internal collapses, because they occur at the same time that the stress falls.

These jumps in the deviator stress are present in realistic experiments of granular soils, but on a smaller scale. In our simulations the magnitude of these fluctuations can be partially attributed to the small size of the sample. One may ask the question if these fluctuations disappear as the size of the sample increases. According to the simulation experiments of shear cells with this model (Tillemans and Herrmann 1995) the answer will be “no”. Indeed, they show that during shear the statistical distribution of the amplitudes of these fluctuation follows a power law dependence, which is qualitatively similar to the seismic activity in the tectonic faults.

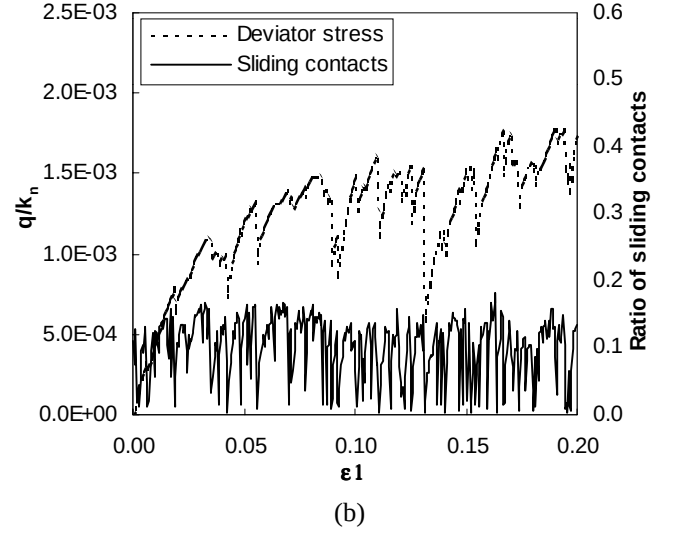
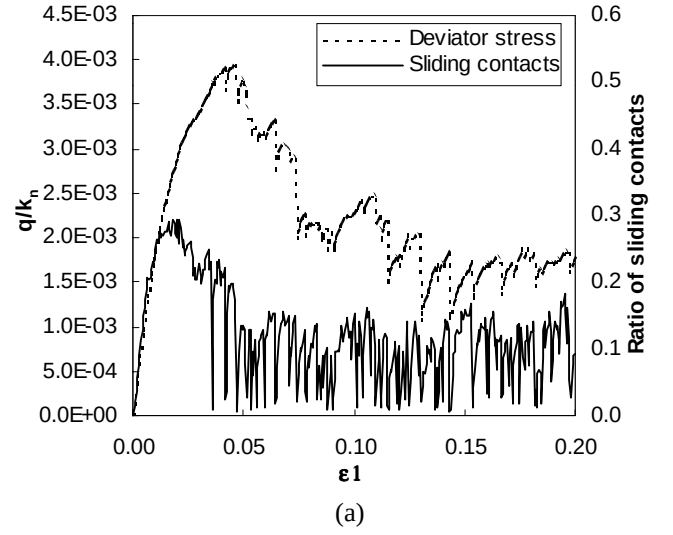


Fig. 6. Evolution of stress-strain and the fraction of sliding contacts with axial strain (a) dense, (b) loose

3.1.3 Displacement field

Strain localization has been experimentally studied by several researchers in the last two decades, e.g. Vardoulakis (1980), Desrues (1984) and others. Additionally, using the discrete element method Cundall (1989), and Bardet and Proubet (1991) have studied this phenomenon. In this paper a brief analysis of strain localization is performed by studying the displacement of the individual particles.

In Figure 7 the displacement vectors of the particles for the dense assembly are presented. At the beginning of the test, the displacements are very small and one can observe approximately a symmetrical deformation around the center of the sample (Fig. 7(a)). As the axial strain increases and before the peak strength is reached a slight tendency to strain localization is observed. After having overcome the peak strength, the particle displacements seem to define independent bodies with different displacement directions (a clearer strain localization). This local-

ization persists as the loading increases, and becomes clearer around 8.9% axial strain (Fig. 7(b)), where two “shear bands” are observed. After peak, these shear bands are not constant in time. This can be mainly due to the boundary conditions, which suppress the shape of bodies that have localization.

The particle displacements of Fig. 7(a) and (b) are taken from two consecutive time intervals, which correspond to an increment of 0.1% of axial strain. Figure 7(c) presents the displacement at 20% axial strain measured from the particle initial position. Here three bodies with different direction of displacement, and the areas, shear bands, defined between them are visible.

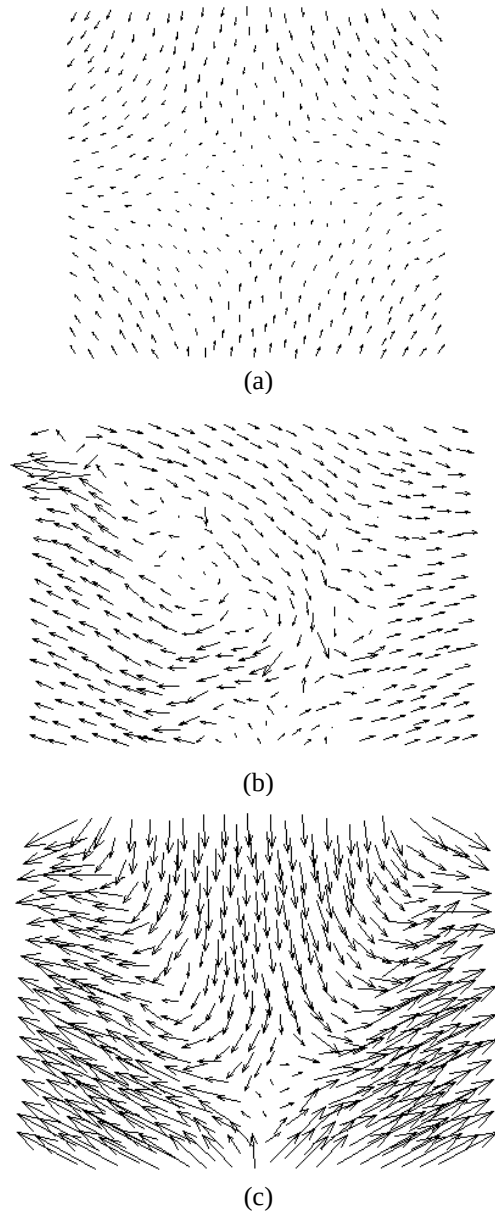


Fig. 7. Displacement of particle centers within a dense sample at three axial strains (a) 1.2%, (b) 8.9% and (c) 20% of total

Although the particle displacements of loose samples are not shown, their evolution reflects the internal structural collapses of the sample, given by large displacements of the particles, associated to the jumps of the stress-strain behavior (Peña 2004). The strain localization is not clear in loose media.

3.1.4 Interparticle friction

An approach to connect the Coulomb friction at the grain level to the macro-mechanical friction is to construct the Mohr-Coulomb failure surface of the granular sample. This failure surface is obtained from the envelope of the Mohr circles at the peak stress value from the biaxial test. The tests were carried out on dense samples, at three confining pressures: 80, 160 and 320 kPa.

Different methods are used to describe the failure surface of granular soils. Linear interpolations are very common, but in some cases power relation are used to take into account the nonlinear variation of the shear strength. Although this nonlinear variation of the shear strength of granular materials with the effective normal stress σ' is widely known, the failure envelope can be approximated by straight segments in small ranges of σ' (Bardet 1997).

In the following analysis as it is used in traditional soil mechanics, we assumed that the failure surface is lineal. Figure 8 shows the failure envelope of a granular sample with interparticle friction coefficient of $\mu = 0.55$ that corresponds to an interparticle friction angle of 28.8° ($\mu = \tan \phi$). The envelope is traced as a tangent straight line to the Mohr circles, and the angle of friction of the bulk material is approximately 39.0° . In this case, the ratio between the friction angle at macro and micro level is equal to 1.35.

In order to study the effect of the interparticle friction ϕ_{micro} on the macro-mechanical friction angle ϕ_{macro} , different interparticle friction coefficients and five different samples were used in the simulations. Additionally to the friction angle at the peak stress, the friction angle at the critical state was also calculated. Figure 9 shows the values of ϕ_{peak} and ϕ_{critic} obtained from variations of ϕ_{micro} . It is observed to low values of ϕ_{micro} that the macro-mechanical angles are similar, that is the granular medium does not mobilize a peak strength. Starting from an interparticle friction angle around 20° the granular samples develop a peak strength ϕ_{peak} . On the contrary, the ϕ_{critic} value seems to remain constant. The last agrees with the experimental results obtained by Skinner (1969), excluding the results from micro-mechanical angle close to zero. These results suggest

for a certain range that the ϕ_{critic} seems to be independent of ϕ_{micro} .

The general trend of the ratio $\phi_{\text{peak}} / \phi_{\text{micro}}$ is as follows: when the interparticle coefficient of friction increases the ratio decreases, and a close value to unitary ratio is obtained for a friction angle of 45° . Values of ϕ_{micro} above 45° exhibit a ratio $\phi_{\text{peak}} / \phi_{\text{micro}}$ lower than one. This is due to strong structural collapses inside the sample that do not allow it to develop the peak strength. When the value of ϕ_{micro} is equal to 0.57, values of ϕ_{peak} around 9.0 are attained. Therefore, the $\phi_{\text{peak}} / \phi_{\text{micro}}$ ratio exhibits a strong increment.

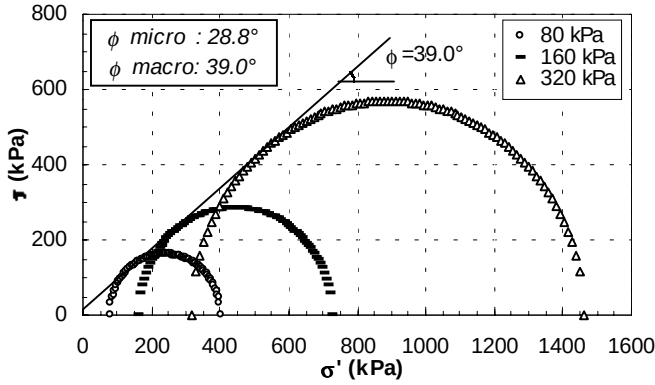


Fig. 8. Mohr-Coulomb failure envelope constructed from biaxial test and Mohr circles

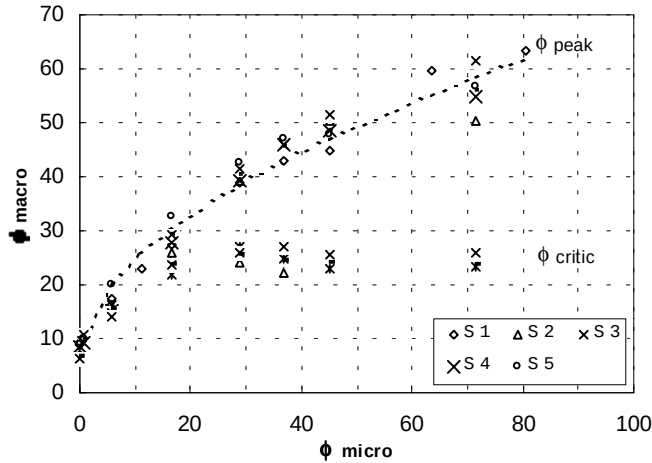


Fig. 9. Evolution of the macro-mechanical friction angle. The solid line is a power law approximation.

Setting ϕ_{micro} to zero, a value of ϕ_{macro} close to 6.0 is obtained. Although this value of ϕ_{macro} is calculated from the average of the fluctuations of the deviator stress, the frictionless granular media offers a small resistance to shear. This result contrasts with the absence of shear resistance, as was found by Thornton (2000) on a system of frictionless spheres. This feature is related to the interlocking among par-

ticles mainly caused by the angularity of their shape. This interlocking upon shear confirms the great influence of the particle shape on the mechanical behavior of the granular media, which is in agreement with the simulation results reported by Mirghasemi et al. (2002), where a bigger angularity of particles is related to a higher shear strength.

Finally, in Eq. (9) we present the power approximation used to fit the data corresponding to the ϕ_{peak} values. This expression considers the effect of the interlocking on the shear strength, and it is only valid for our simulation results.

$$\phi_{\text{peak}} = 5.5 \cdot \phi_{\text{micro}}^{0.53} + 6 \quad (9)$$

3.2 Test 2: Proportional strain paths simulations

The main purpose of this type of test simulation is to confirm if the virtual granular sample reproduces the so-called Swept Out of Memory (SOM) behavior of soils. This kind of behavior has been experimentally observed by Goldscheider (1976 and 1982) and later by Chu, J. and Lo, S.C (1994), in which proportional strain paths starting from a nearly stress free state results in proportional stress paths.

The SOM behavior is an important asymptotic property of granular materials, the same as the critical state. The SOM-hypothesis was posed by Gudehus (1977), and used to state that the grain skeleton is fully described by the stress tensor and the void ratio (Gudehus et al. 1977). This behavior is also independent of the initial stress state (Gudehus 1997, Herle and Gudehus 1999, Kolymbas 2000). For example, the obtained stress paths corresponding to a not stress free initial state, converge asymptotically to the stress path related to a stress free state, as is shown in Figure 11.

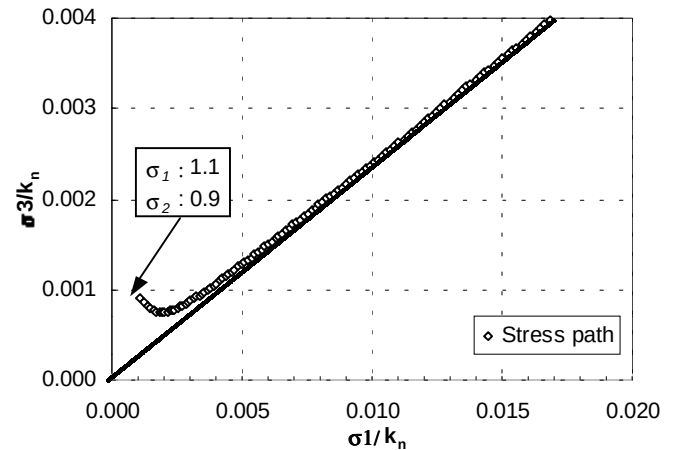


Fig. 11. Proportional stress path obtained from numerical simulation.

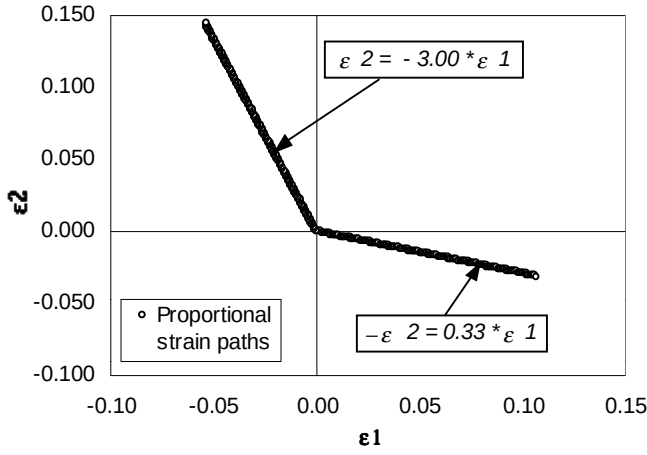


Fig. 12. Proportional strain paths used to perform the numerical simulations.

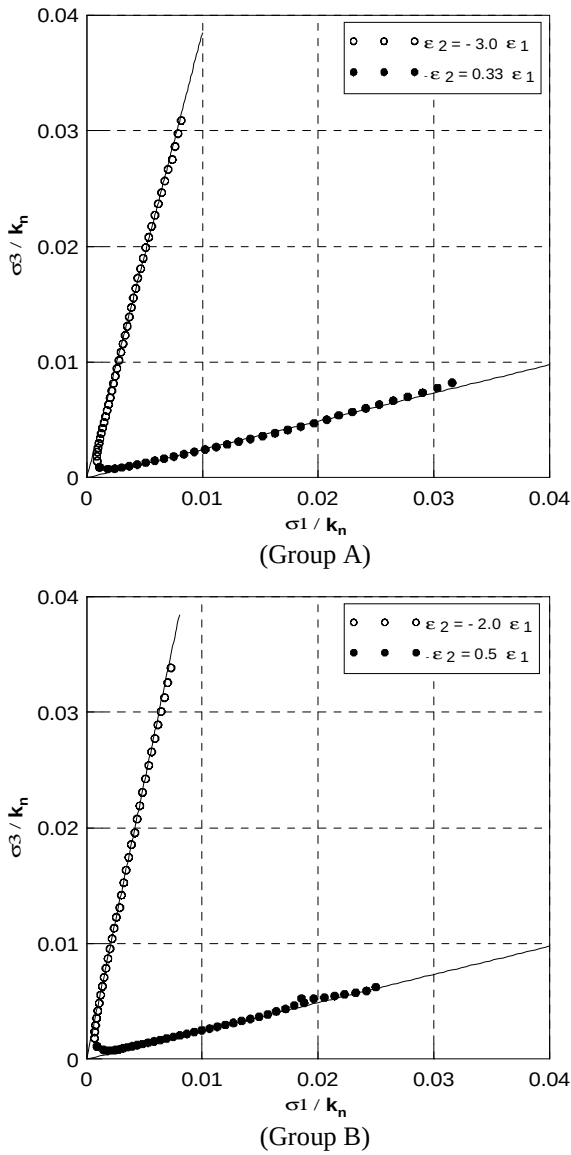


Fig. 13. Proportional stress paths corresponding to different proportional strain paths in numerical simulations.

With the aim of verifying the convergence of the granular samples to the Swept Out of Memory behavior, different strain paths and two initial stress states were used to carry out numerical simulations. These parameters are given in Table 5. In Figure 12, two proportional strain paths used to perform the tests are presented.

Group	Initial stress state / k_n	Proportional strain path
A	$\sigma_1 = 1.0$	$\varepsilon_2 = -3.0 \varepsilon_1$
	$\sigma_2 = 0.9$	$-\varepsilon_2 = 0.33 \varepsilon_1$
B	$\sigma_1 = 0.9$	$\varepsilon_2 = -2.0 \varepsilon_1$
	$\sigma_2 = 1.0$	$-\varepsilon_2 = 0.5 \varepsilon_1$

Finally, Figure 13 shows results of the proportional strain paths test performed. Although a slight deviation is observed in one of the samples, the general tendency of the virtual granular medium is to approach asymptotically the path starting from the stress free state. In agreement with the Swept Out of Memory behavior of soils.

4 CONCLUSIONS

In this paper the effect of the initial density of the sample and the interparticle coefficient of friction on the macro-mechanical behavior of granular materials was investigated. The results show that at large strains the samples reach the critical state independent on their initial density, and they deform at constant void ratio, volume and mechanical coordination number. The evolution of the deviator stress exhibits fluctuations around the residual strength. These fluctuations are related to the fraction of sliding contacts, but it is also important to study different aspects that can induce this behavior, such as the boundary conditions, strain rate, and others. Studying the displacement of the individual particles strain localization is observed within the granular sample. Thus, during the shear process bodies with different displacement direction and shear bands could be identified.

The dependency of the macro-mechanical friction angle at the peak strength on the interparticle friction coefficient can be approximately fitted by $\phi_{peak} = 5.5 \cdot \phi_{micro}^{0.53} + 6$. The intercept value of 6.0 in this relation shows the effect of the interlocking between particles due to their angularity. The relation between the friction angle at the critical state and the interparticle friction coefficient shows that it becomes independent of ϕ_{micro} , when ϕ_{micro} is greater than 18 degrees. This last suggests that the critical state of the granular material under mono-

tonic loading may be independent of the interparticle friction coefficient.

Performing proportional strain paths tests was found that these strain paths are related to proportional stress paths. It would be important to verify if samples with different initial condition converge to the same asymptotic behavior, which is one of the most important hypothesis of the hypoplastic theory of the continuum.

Although numerical simulations have several restrictions (sample size, possible kinematic constraints for being a system in two dimensions), they are a valuable tool for modeling soil behavior. Moreover, contrary to the constitutive models of the continuum, our numerical simulations only need two parameters: the interparticle friction coefficient and the stiffness ratio at the contact. Future work can be focused on the influence that variations in the grain size distribution have on the mechanical behavior of the granular media. Furthermore, they can include crushing of particles, and introduce pore pressure.

5 REFERENCES

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